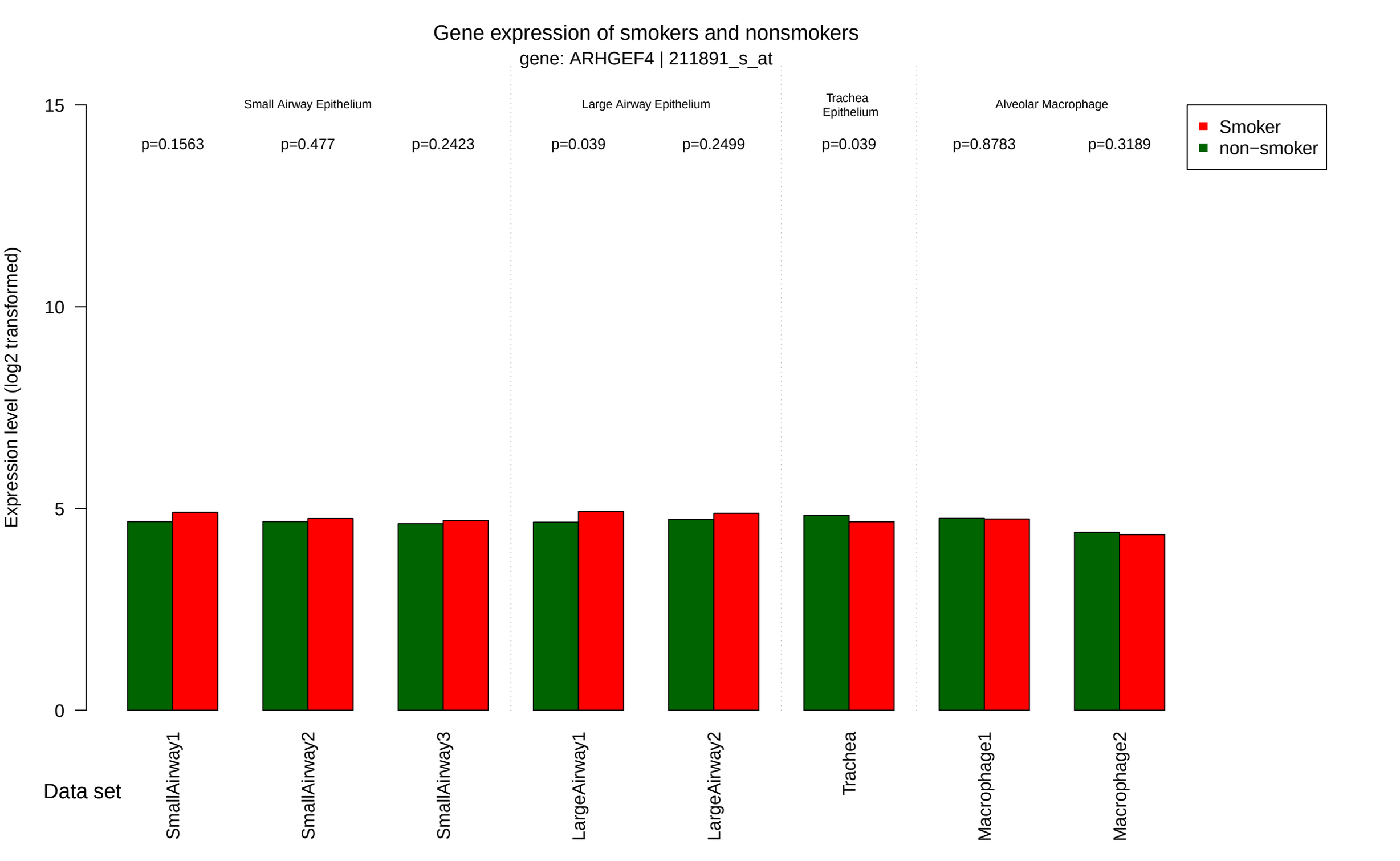
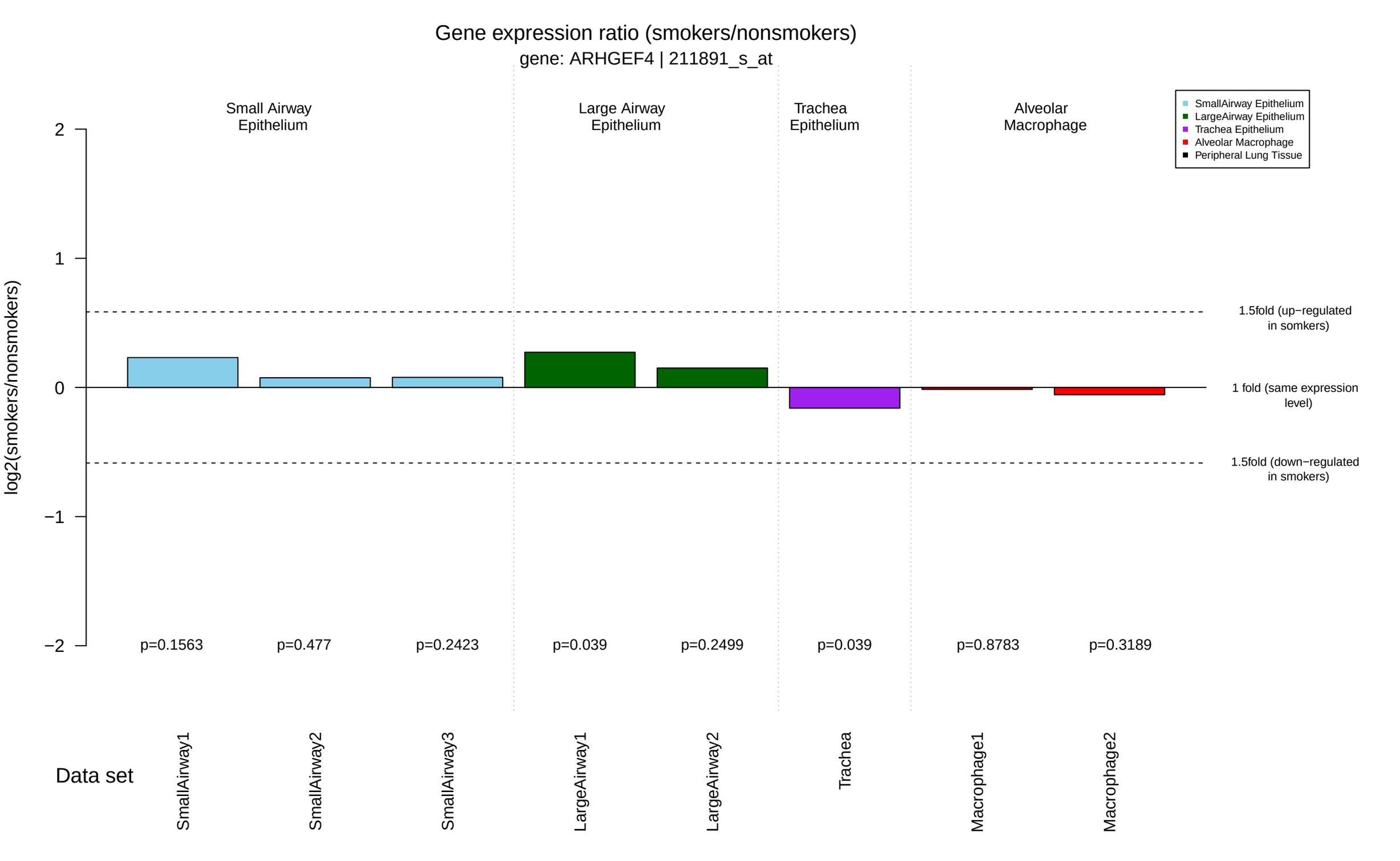
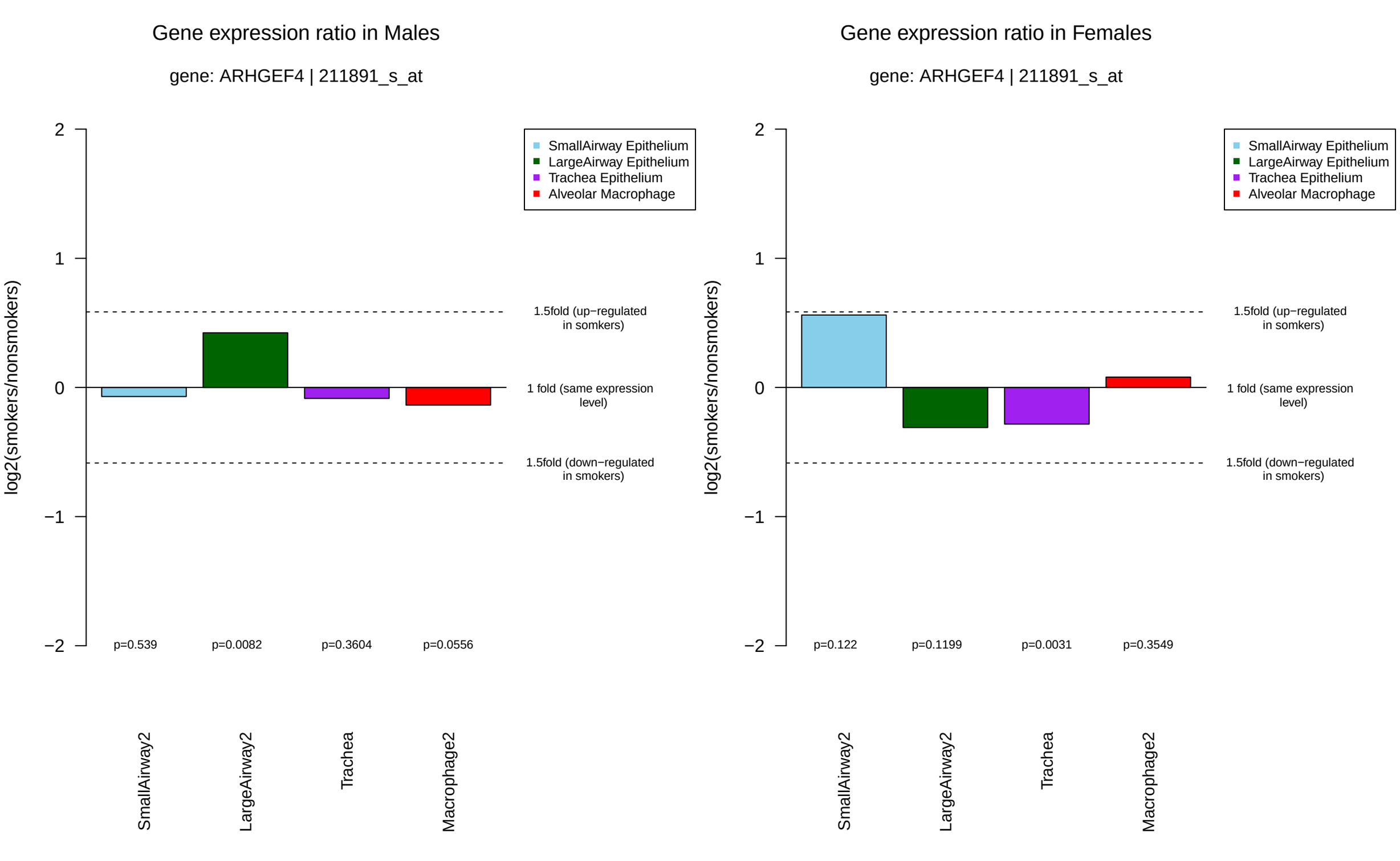
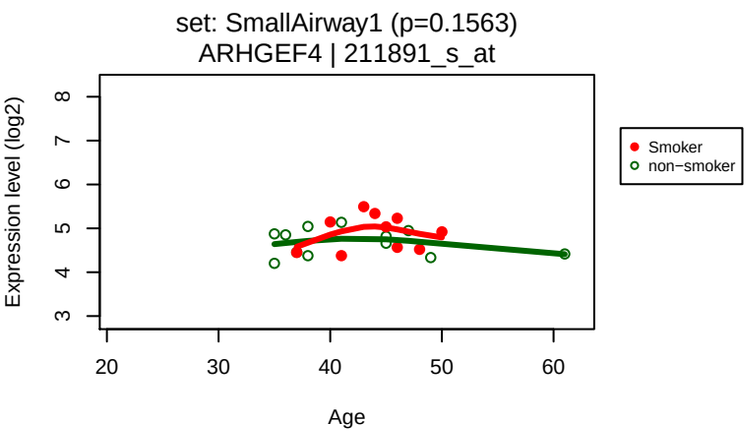
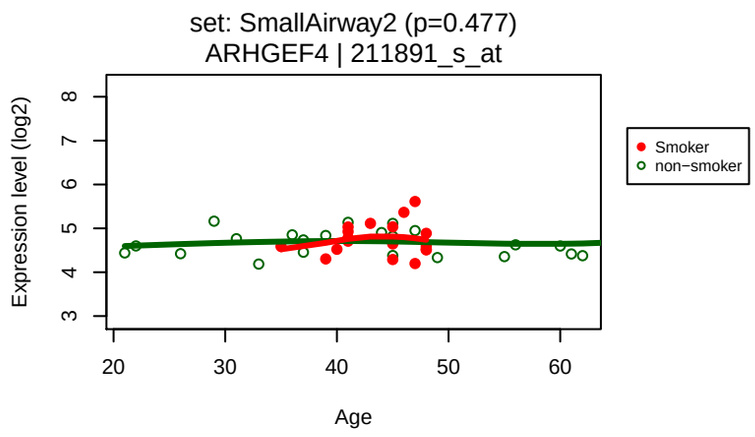
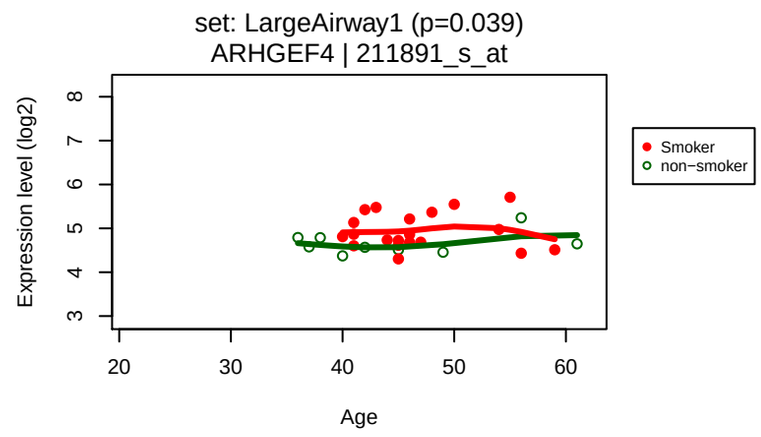

Gene expression of smokers and nonsmokers
gene: ARHGEF4 | 211891_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1563
p=0.477
p=0.2423
p=0.039
p=0.2499
p=0.039
p=0.8783
p=0.3189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

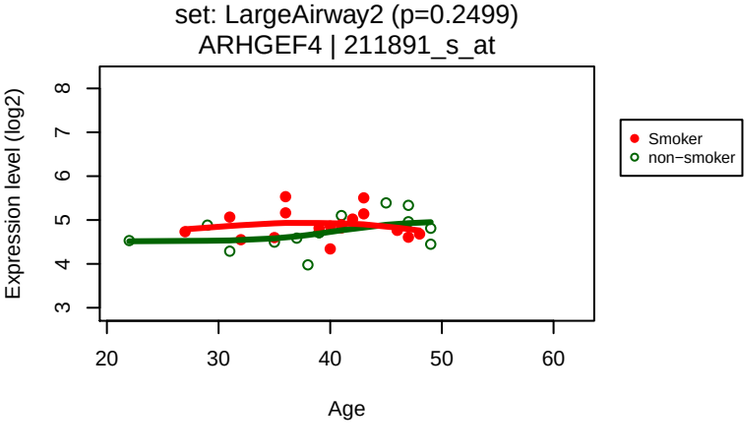

Gene expression ratio (smokers/nonsmokers)
gene: ARHGEF4 | 211891_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1563
p=0.477
p=0.2423
p=0.039
p=0.2499
p=0.039
p=0.8783
p=0.3189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

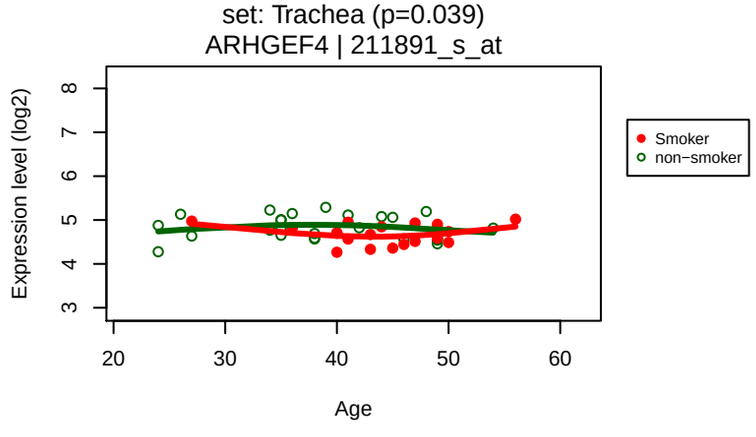

Gene expression ratio in Males
gene: ARHGEF4 | 211891_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.539
p=0.0082
p=0.3604
p=0.0556
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGEF4 | 211891_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.122
p=0.1199
p=0.0031
p=0.3549
SmallAirway2
LargeAirway2
Trachea
Macrophage2

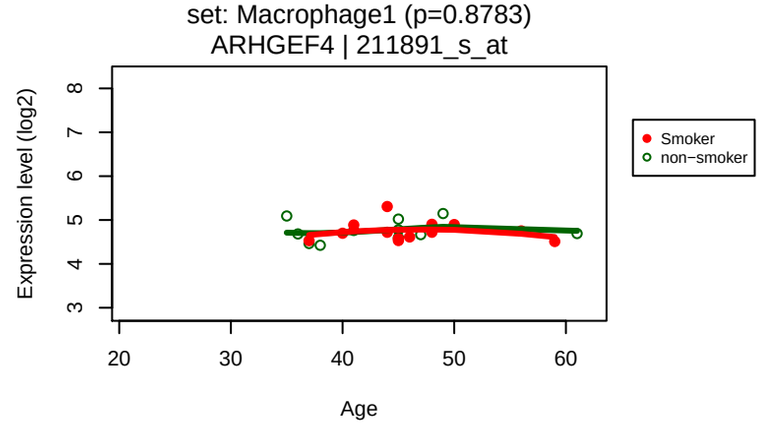

set: SmallAirway1 (p=0.1563)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker

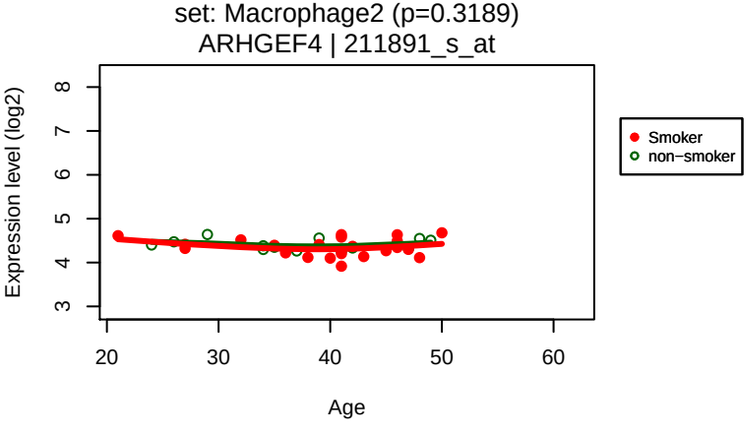

set: SmallAirway2 (p=0.477)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker

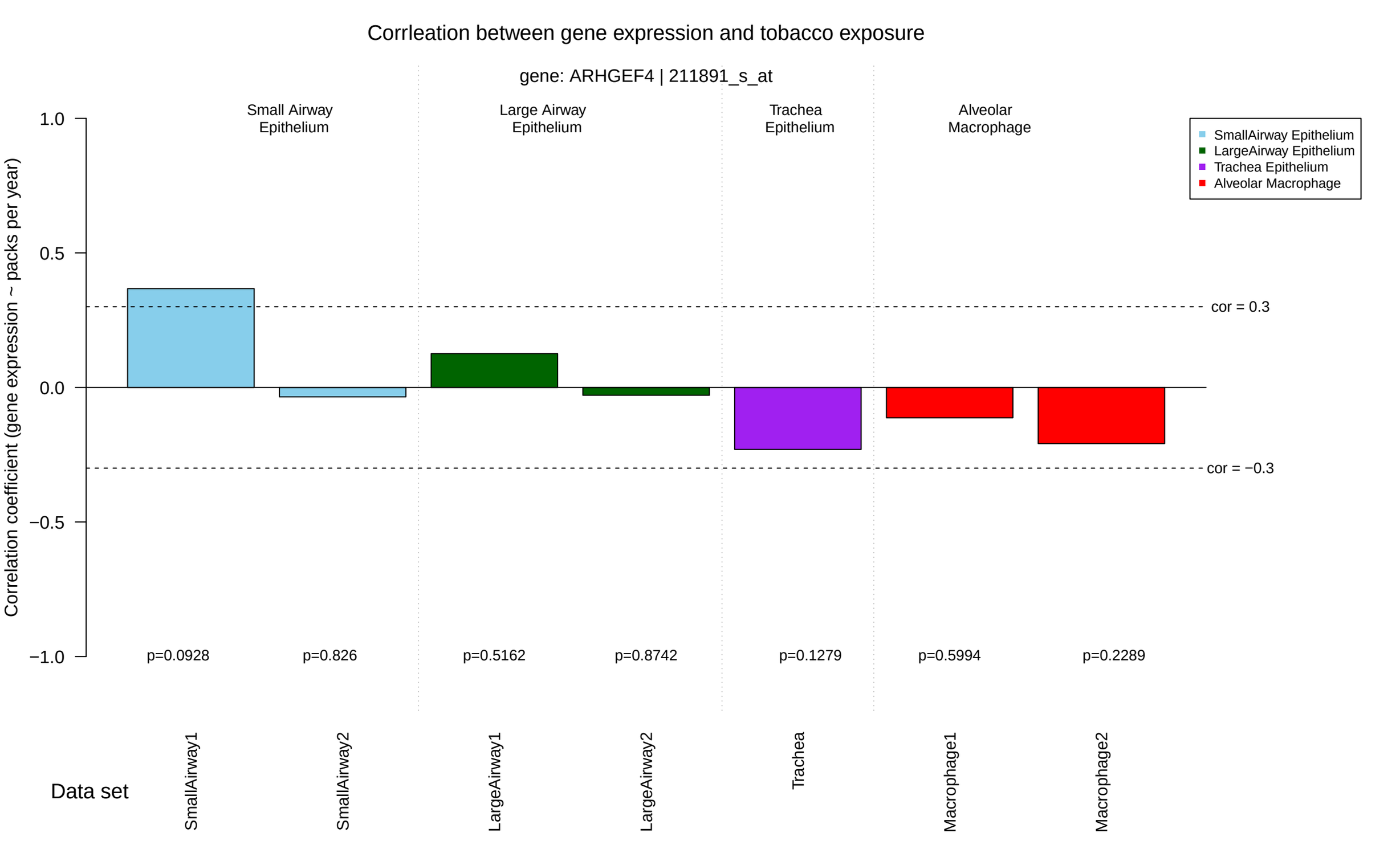

set: LargeAirway1 (p=0.039)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2499)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.039)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8783)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3189)
ARHGEF4 | 211891_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGEF4 | 211891_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0928
p=0.826
p=0.5162
p=0.8742
p=0.1279
p=0.5994
p=0.2289
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2